ModelIt!

Teacher's Guide

Why Baking Soda Explodes: Reaction Rates and Energy

How Reactant Amount and Surface Area Control Chemical Reactions

Quick Reference

Grade Level	Grade 08L2
NGSS Standard	MS-PS1-2, MS-PS1-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-PS1-2, MS-PS1-5

Analyze and interpret data on the properties of substances before and after the substances interact to determine if a chemical reaction has occurred. Develop and use a model to describe how the total number of atoms does not change in a chemical reaction and thus mass is conserved.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model showing how reactant amount and surface area control reaction rate and demonstrate conservation of mass.
Disciplinary Core Idea	**PS1.B Chemical Reactions** — The total number of atoms does not change during a chemical reaction; atoms rearrange to form new substances while conserving mass.
Crosscutting Concept	**Energy and Matter** — Students track matter through a chemical reaction, demonstrating that atoms are conserved even as they form new arrangements and new substances.

CAST Testing Alignment

MULTIPLE CHOICE:

A student seals baking soda and vinegar in a plastic bag, weighs it, lets the reaction occur, and weighs it again. The mass stays the same. Then the student opens the bag and weighs it. The mass decreases. Which best explains this?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain how both reactant amount and surface area affect reaction rate. Include a description of what happens to total system mass during the reaction and why.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Chemical reactions involve the rearrangement of atoms from reactant molecules into product molecules. Understanding what controls reaction rate and how mass is conserved is fundamental to chemistry, engineering, and everyday life — from cooking to manufacturing to medicine.

The Baking Soda-Vinegar Reaction

When sodium bicarbonate (baking soda, NaHCO3) reacts with acetic acid (vinegar, CH3COOH), three products form: sodium acetate (NaCH3COO), water (H2O), and carbon dioxide gas (CO2). The fizzing and bubbling is CO2 escaping. Every atom from the reactants ends up in the products — nothing is created or destroyed.

Reaction Rate Factors

Reaction rate depends on how often reactant particles collide with enough energy to react. Surface area increases the number of exposed particles available for collision. More reactant means more particles overall. Higher temperature increases particle speed and collision frequency. Catalysts lower the energy barrier for successful collisions.

Conservation of Mass

Antoine Lavoisier established the law of conservation of mass in the 1780s. In any chemical reaction, the total mass of reactants equals the total mass of products. When reactions seem to lose mass (like burning wood), the 'missing' mass has become gases that escaped into the atmosphere. In sealed systems, mass remains constant.

Surface Area and Particle Size

Grinding a solid into powder dramatically increases its surface area. A 1 cm cube has 6 cm^2 of surface area, but crushing it into powder can produce hundreds of cm^2 of surface area from the same volume. This is why powdered sugar dissolves faster than sugar cubes, and why grain dust in silos can explode — maximized surface area means maximized reaction rate.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify reactant amount, surface area, reaction rate, gas production, and total system mass as system components.

E **Establish Relationships**
Students determine that reactant amount and surface area both positively affect reaction rate, which drives gas production, while total system mass remains constant.

V **Visualize & Model**
Students build the chemical reaction model showing how inputs control rate and outputs while mass is conserved.

E **Evaluate Outcomes**
Students compare maximum explosion vs. slow fizz scenarios and verify that total system mass is identical in both cases.

R **Revise & Extend**
Students add temperature or catalyst components to explore additional factors affecting reaction rate.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Baking soda and vinegar — you've done it since kindergarten. But do you know WHY it explodes?"*

Watch For

Quick demonstration: baking soda + vinegar in a flask. Build curiosity.

Slide 2: Learning Objectives

3 min

Say: *"Today we go deeper — modeling the science behind every fizzy reaction."*

Watch For

Have students read objectives. Pre-teach 'conservation of mass.'

Slide 3: Big Question

3 min

Say: *"If gas escapes during a reaction, is mass really conserved? Let's find out."*

Watch For

Weigh a sealed bag before and after the reaction. Reveal the answer later.

Slide 4: LEVER Framework

2 min

Say: *"We'll model what controls how fast and how violently this reaction happens."*

Watch For

Preview the connection between surface area and explosion intensity.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of this chemical system? What can we change?"*

Watch For

Guide sorting. Discuss why surface area matters even when the amount is the same.

Slide 6: Activity 2: Connections

8 min

Say: *"More surface area means more collisions between reactant particles — faster reaction!"*

Watch For

Map relationships. Use collision analogy (dodgeball with more players).

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's compare: powder explodes while chunks barely fizz. But is the total mass different?"*

Watch For

Students predict, then run. Focus on the Total System Mass line staying flat.

Slide 8: Discoveries

5 min

Say: *"The gas has mass! When you seal the system, nothing is lost. Atoms rearrange, not disappear."*

Watch For

Return to the sealed bag demonstration. Weigh it again. Mass unchanged!

Slide 9: STEM Challenge

12 min

Say: *"Prove to the skeptics: mass is conserved. Design an experiment that shows it."*

Watch For

Teams design sealed reaction experiments. Present evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Reactant Amount, Surface Area
- **INTERNAL:** Reaction Rate, Gas Production (CO_2), Total System Mass

Relationships

- **Reactant Amount to Reaction Rate:** POSITIVE (+) — More reactant means more particles available to collide and react, increasing the overall rate of the reaction.
- **Surface Area to Reaction Rate:** POSITIVE (+) — Greater surface area exposes more reactant particles for contact, dramatically increasing collision frequency and reaction speed.
- **Reaction Rate to Gas Production:** POSITIVE (+) — A faster reaction produces carbon dioxide gas more quickly, creating more vigorous bubbling and fizzing.
- **Reaction Rate to Reactant Amount:** NEGATIVE (-) — As the reaction proceeds, reactants are consumed and converted into products, decreasing the amount of reactant remaining.
- **Total System Mass remains CONSTANT:** NEUTRAL (=) — In a sealed system, the total mass of all reactants equals the total mass of all products. Atoms are rearranged, not created or destroyed.

Key Discoveries

- More reactant produces more product, but at the same rate per unit of reactant
- Increasing surface area dramatically speeds up the reaction — powder reacts faster than chunks
- Total System Mass stays constant in a sealed system, even though gas is produced
- When the container is open, mass appears to decrease because gas escapes — but the gas still has mass
- Reactants are consumed as products form — the lines on the graph are inverse mirrors

Multiple Choice Post-Assessment Answer Key

1. In your model, both the 'Maximum Explosion' and 'Slow Fizz' scenarios produced the same final amount of CO_2 gas. Which principle does this evidence demonstrate?

Correct: B) The amount of product depends on the amount of reactant, not the speed of the reaction

Correct! The model shows that reaction rate (speed) and total product (amount) are different things. Surface area affects how fast the reaction proceeds, but the total CO_2 produced depends on how much baking soda and vinegar reacted. Same reactant amount means same product amount.

2. A student weighs a sealed bag containing baking soda and vinegar before and after the reaction. The mass stays the same. Then the student opens the bag, and the mass decreases. Which explanation is supported by the model?

Correct: B) CO_2 gas produced during the reaction has mass and escaped when the bag was opened, but no atoms were created or destroyed

Correct! The model demonstrates conservation of mass. In the sealed bag, total mass was constant because all products (including CO_2 gas) remained inside. Opening the bag allowed CO_2 to escape, carrying its mass with it. No atoms were created or destroyed.

3. Your model shows that increasing Surface Area from 20% to 100% dramatically increased Reaction Rate, even though Reactant Amount stayed the same. What particle-level explanation supports this?

Correct: B) Powder exposes more reactant particles for contact with vinegar, increasing collision frequency and reaction speed

Correct! Grinding baking soda into powder increases the surface area exposed to vinegar without changing the total amount. More exposed particles means more simultaneous collisions between reactants, dramatically increasing reaction speed.

4. The model shows a NEGATIVE relationship from Reaction Rate back to Reactant Amount. What does this represent in the chemical reaction?

Correct: C) As the reaction proceeds, reactants are consumed and converted into products, decreasing the remaining reactant

Correct! This negative feedback represents reactant consumption. As baking soda and vinegar react, they are converted into sodium acetate, water, and CO2. The reactants are used up as products form. The atoms are rearranged, not destroyed.

5. Based on your model, a student wants to design an experiment that produces the most intense and fastest reaction possible. Which combination of model inputs should they use, and why?

Correct: C) High Reactant Amount and high Surface Area, because both inputs positively affect Reaction Rate

Correct! The model shows that both Reactant Amount and Surface Area have POSITIVE relationships with Reaction Rate. Maximizing both inputs produces the most particles available (high amount) with the most exposed surface for collisions (high surface area), creating the fastest, most intense reaction.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Gas produced means mass is lost"	The gas (CO_2) has mass just like any other substance. In an open container, the gas escapes and takes its mass with it — but the atoms still exist. In a sealed container, total mass stays exactly the same.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Chemical reactions create new atoms"	Chemical reactions REARRANGE existing atoms into new combinations. No atoms are created or destroyed. The same carbon, hydrogen, oxygen, and sodium atoms from the reactants end up in the products — just in different molecular arrangements.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Faster reactions produce more product"	Faster reactions produce the SAME amount of product — just more quickly. The total product depends on the amount of reactant, not the speed of the reaction. A powder explosion and a chunk fizz produce the same total CO_2 from the same mass of baking soda.